

TAHLEQUAH, OK, USA

WMO: 722199

Lat: 35.929N

Lon: 95.004W

Elev: 266

StdP: 98.17

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 53956

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.1	-7.9	-16.0	1.0	-7.6	-12.8	1.3	-5.5	10.6	7.9	9.2	7.6	1.7	330	0.436

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.9	37.4	25.5	36.0	25.1	33.8	24.5	26.9	34.0	26.2	33.4	25.5	32.5	2.8	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.2	21.0	30.6	24.1	19.6	29.2	23.6	19.1	28.8	86.9	34.9	82.8	33.4	79.4	32.6	29.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.8	7.8	7.0	-15.9	39.5	3.6	2.3	-18.5	41.1	-20.7	42.5	-22.7	43.7	-25.4	45.4		
(5)			-16.4	27.6	3.6	1.1	-18.9	28.4	-21.0	29.0	-23.0	29.6	-25.6	30.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	15.3	3.0	5.1	10.5	15.0	19.7	24.7	26.9	26.5	22.3	15.4	9.3	4.2	(6)
(7)		DBStd	9.68	5.72	6.18	5.77	4.84	4.40	2.73	2.86	3.18	3.75	5.16	5.36	5.35	(7)
(8)		HDD10.0	744	226	160	64	12	2	0	0	0	0	12	75	193	(8)
(9)		HDD18.3	2096	475	371	249	120	38	1	0	0	10	121	272	439	(9)
(10)		CDD10.0	2668	10	22	80	162	302	441	523	512	370	178	56	13	(10)
(11)		CDD18.3	978	0	1	7	20	80	191	266	254	129	29	2	0	(11)
(12)		CDH23.3	10491	1	4	53	194	750	1996	3097	2818	1280	277	19	0	(12)
(13)		CDH26.7	4934	0	1	6	35	223	879	1664	1499	553	72	2	0	(13)
(14)	Wind	WSAvg	2.4	2.6	2.9	3.2	3.2	2.5	2.2	1.7	1.7	1.8	2.1	2.5	2.4	(14)
(15)	Precipitation	PrecAvg	1200	58	61	99	108	144	126	95	88	123	106	101	76	(15)
(16)		PrecMax	1866	169	194	247	212	307	361	341	244	322	314	288	191	(16)
(17)		PrecMin	654	2	4	14	6	41	23	3	0	7	3	10	7	(17)
(18)		PrecStd	250	43	38	54	49	64	72	80	58	79	80	65	52	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.3	23.0	27.4	30.8	32.7	36.4	39.1	40.9	36.1	31.3	26.2	22.0	(19)
(20)			MCWB	14.5	14.7	17.8	19.1	23.8	25.4	25.2	25.7	23.4	21.8	18.3	16.8	(20)
(21)		2%	DB	18.0	21.1	24.9	27.4	31.2	34.0	37.4	37.7	33.0	28.6	22.7	17.9	(21)
(22)			MCWB	13.0	14.6	16.9	18.7	23.1	24.7	25.7	25.5	23.3	21.2	15.9	13.8	(22)
(23)		5%	DB	16.1	17.8	22.4	26.0	29.0	32.7	36.1	36.1	32.2	27.0	21.0	16.1	(23)
(24)			MCWB	10.7	12.8	15.6	17.7	22.2	24.4	25.5	24.9	22.9	19.2	15.7	11.7	(24)
(25)		10%	DB	12.7	16.0	20.3	23.6	27.5	32.0	33.9	33.1	30.2	24.0	18.1	12.9	(25)
(26)			MCWB	8.0	11.4	14.5	17.0	21.0	24.3	24.9	24.3	21.9	18.3	13.5	9.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.9	17.6	19.6	22.2	25.6	27.2	28.4	28.0	25.7	23.6	19.7	17.5	(27)
(28)			MCDWB	19.3	21.5	25.5	27.2	31.1	33.2	35.2	35.5	31.4	29.0	24.2	20.4	(28)
(29)		2%	WB	14.8	15.7	18.4	20.3	24.3	26.2	27.3	26.7	24.5	22.2	18.1	15.3	(29)
(30)			MCDWB	17.7	19.2	23.2	25.2	28.8	32.7	34.1	34.6	30.6	26.4	20.8	17.4	(30)
(31)		5%	WB	11.4	13.9	17.1	19.2	23.2	25.4	26.3	25.8	23.7	20.9	16.2	12.8	(31)
(32)			MCDWB	15.0	16.9	21.0	23.6	27.6	31.7	33.6	33.6	29.7	23.7	19.5	14.9	(32)
(33)		10%	WB	8.8	11.6	15.5	18.0	22.1	24.6	25.6	25.2	22.9	19.6	14.7	10.5	(33)
(34)			MCDWB	12.7	15.4	19.2	22.0	26.3	30.1	32.7	32.4	28.5	23.0	18.1	12.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	12.3	12.2	12.9	13.4	12.3	11.8	12.9	13.0	13.3	13.6	13.6	11.7	(35)
(36)			MCDBR	16.5	15.3	14.8	15.1	13.8	13.6	15.1	15.2	14.9	15.3	14.8	15.2	(36)
(37)			MCWBR	11.1	10.0	8.6	8.3	6.8	5.8	5.2	5.0	5.7	7.6	9.2	10.5	(37)
(38)		5% WB	MCDWB	14.1	13.2	12.6	11.7	11.8	12.4	13.1	13.4	12.6	10.9	12.7	12.2	(38)
(39)			MCWBR	10.7	9.8	8.1	7.5	6.6	6.0	5.5	5.2	5.6	6.2	9.4	9.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.305	0.310	0.350	0.385	0.424	0.427	0.445	0.446	0.393	0.349	0.317	0.297	(40)	
(41)		taud	2.487	2.486	2.405	2.317	2.266	2.283	2.281	2.251	2.368	2.476	2.515	2.525	(41)	
(42)		Ebn at Noon	897	934	919	898	864	857	839	831	862	877	875	885	(42)	
(43)		Edh at Noon	83	93	110	127	136	133	133	134	113	93	80	75	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.51	3.04	4.07	5.09	5.64	6.44	6.47	5.88	4.96	3.71	2.69	2.14	(44)	
(45)		RadStd	0.24	0.44	0.25	0.30	0.51	0.64	0.40	0.39	0.38	0.49	0.32	0.27	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (33 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page